PARVA BAROT

Software Engineer
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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building scalable backend systems, distributed applications, cloud-native services, and AI-enabled enterprise platforms. Proficient in Python, Java, FastAPI, Spring Boot, AWS, Docker, and Kubernetes, with experience supporting production workloads exceeding 500K daily requests and 1M asynchronous events. Strong background in REST APIs, microservices, CI/CD, performance optimization, observability, infrastructure automation, and data processing, with hands-on experience integrating LLM, RAG, vector search, and intelligent retrieval capabilities into production software systems.

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, JavaScript, SQL
Backend Engineering: FastAPI, Django, Spring Boot, .NET Core, REST APIs, gRPC, Microservices, API Design
Distributed Systems & Messaging: Apache Kafka, Event-Driven Architecture, Asynchronous Processing, Distributed Systems
Cloud & DevOps: AWS (EC2, S3, Lambda, EKS, SQS), Docker, Kubernetes, Terraform, Helm, GitHub Actions, CI/CD
Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis, TimescaleDB, Pinecone, ChromaDB
Software Quality & Observability: Unit Testing, Integration Testing, Test Automation, Logging, Monitoring, Performance Tuning, Incident Triage
AI/ML: LLMs, Retrieval-Augmented Generation (RAG), LangChain, OpenAI API, Vector Databases, AI Agents, LangGraph, Embeddings, Semantic Search

PROFESSIONAL EXPERIENCE

Citigroup, AZ | Software Engineer | November 2024 – Present

- Designed and developed scalable backend systems and REST APIs using Python and Java for enterprise platforms, supporting 500K+ daily requests while improving platform reliability and request throughput.
- Built high-throughput microservices using FastAPI, Docker, and Kubernetes, reducing p95 API response latency by 28% under high-concurrency workloads.
- Designed scalable cloud-native services and distributed processing infrastructure supporting 1M+ asynchronous events per day while improving workload scalability and production reliability.
- Implemented CI/CD pipelines with automated testing, deployment automation, monitoring, telemetry, and logging for production applications, reducing average deployment time by 40%.
- Architected and integrated AI-powered services, including LLM and RAG capabilities, into five enterprise software platforms and business workflows.
- Built scalable retrieval, caching, and data-access systems supporting 75K+ indexed documents using Redis, vector search, and distributed data-processing technologies, reducing p95 retrieval latency by 30%.

Infinite Infolab, India | Software Engineer | February 2021 – July 2023

- Designed and developed scalable backend systems, REST APIs, and microservices in Python and Java, improving application performance, reliability, and p95 response time by 30%.
- Built asynchronous data and document-ingestion pipelines using FastAPI and AWS Lambda, processing up to 10K documents per day for scalable processing and workflow automation.
- Developed intelligent search, retrieval, and data-access capabilities across 500K+ indexed documents using Redis and vector databases, reducing average search latency by 25%.
- Integrated frontend and backend services to support real-time workflows, API-driven applications, and automated document processing, reducing average workflow completion time by 25%.
- Containerized and optimized microservices using Docker multi-stage builds, reducing container image size by 35% while improving deployment consistency, scalability, portability, and runtime efficiency.
- Collaborated with cross-functional teams to deploy secure cloud-native applications with authentication, monitoring, logging, observability, and scalable infrastructure, reducing production incident investigation time by 20%.

EDUCATION

ARIZONA STATE UNIVERSITY, Tempe, AZ, USA
Master of Science in Information Technology | August 2023 – May 2025 | GPA: 4.0 / 4.0

SELECTED PROJECTS

Retrieval-Augmented Generation Platform | Python, FastAPI, Redis, Docker, Kubernetes

- Built a scalable semantic search platform supporting 50K+ documents using Redis Vector Search, LangChain, and vector databases.
- Developed streaming and low-latency APIs with FastAPI and deployed containerized services using Docker and Kubernetes.
- Reduced retrieval latency by 35% through embedding optimization, intelligent caching, and performance tuning.

Multi-Agent Workflow Automation System | Python, GPT-4, LangGraph, Kafka, AWS

- Designed event-driven workflow orchestration pipelines for automated document classification and asynchronous task execution.
- Integrated AI services with Apache Kafka architecture for scalable asynchronous processing.
- Improved task completion accuracy by 22% using contextual memory management and prompt optimization.

LLM Evaluation & Monitoring Platform | Python, MLflow, Prometheus

- Built evaluation pipelines for hallucination detection, latency tracking, and response scoring across production workloads.
- Implemented telemetry dashboards and automated benchmark reporting for system reliability and performance monitoring.